Lab 2: Finite Difference Approximation Applied to the Heat Equation

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Excel was used to solve both problems.

Steady-State Heat Conduction

For steady-state heat conduction, our object, a square column, is 100℃ on one face and 0℃ on the other three. We then use the following formula to investigate the steady-state temperature at some $t > 0$.

$$T_{m,n}^{\text{new}} = \frac{\dfrac{T_{m+1,n}^{\text{old}} + T_{m-1,n}^{\text{old}}}{(\Delta x)^2} + \dfrac{T_{m,n+1}^{\text{old}} + T_{m,n-1}^{\text{old}}}{(\Delta y)^2}}{\dfrac{2}{(\Delta x)^2} + \dfrac{2}{(\Delta y)^2}}$$

The system stabilized after some iterations and gave us the graph seen in Figure 1.

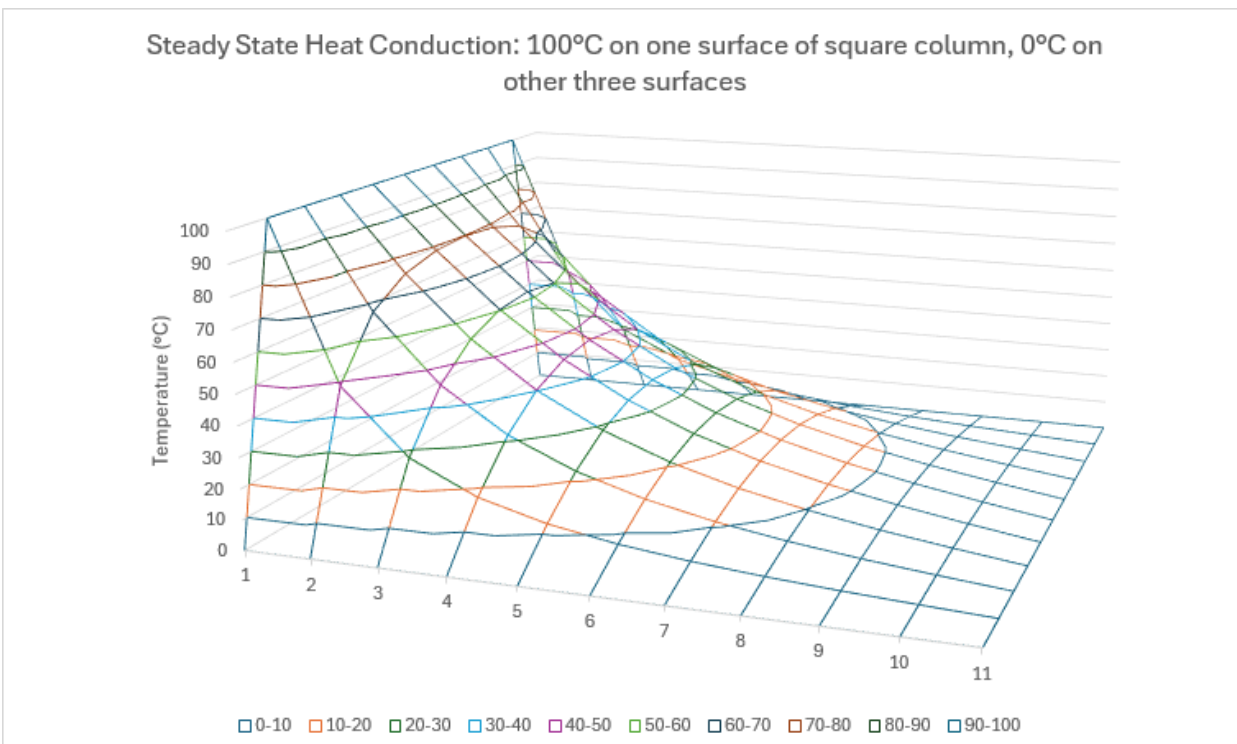

Figure 1: Steady state of a square column with one 100℃ face and three 0℃ faces.

Transient State

For the transient state, we simulated how heat will transfer within a square column at 100℃ when it is submerged in ice water (0℃). The following formula was used.

$$T_{m,n}^{p+1} = T_{m,n}^{p} + \alpha \Delta t \left(\frac{T_{m+1,n}^{p} + T_{m-1,n}^{p} - 2T_{m,n}^{p}}{(\Delta x)^2} + \frac{T_{m,n+1}^{p} + T_{m,n-1}^{p} - 2T_{m,n}^{p}}{(\Delta y)^2} \right)$$

with additional constraint defined as,

$$1 - \frac{2\alpha\Delta t}{(\Delta x)^2} - \frac{2\alpha\Delta t}{(\Delta y)^2} > 0$$

With the constraint, we chose $\Delta x = 1; \Delta y = 1, \Delta t = 1$, and $\alpha = 0.062$. This is the initial state of the square column

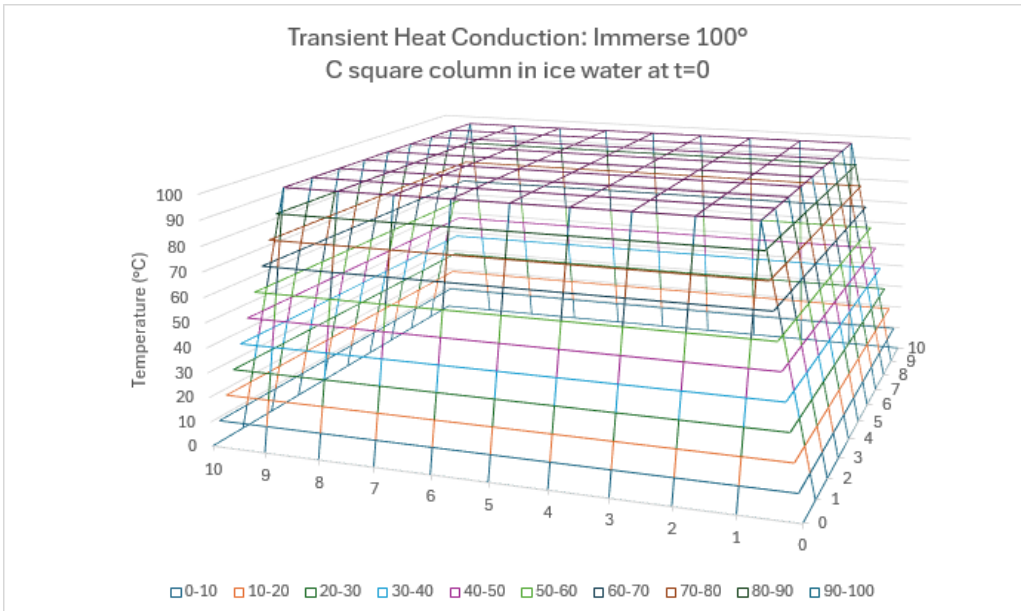

Figure 2: Transient state of square column at 100°C submerged in 0°C ice water at t = 0

This is the transient state at $t = 12$

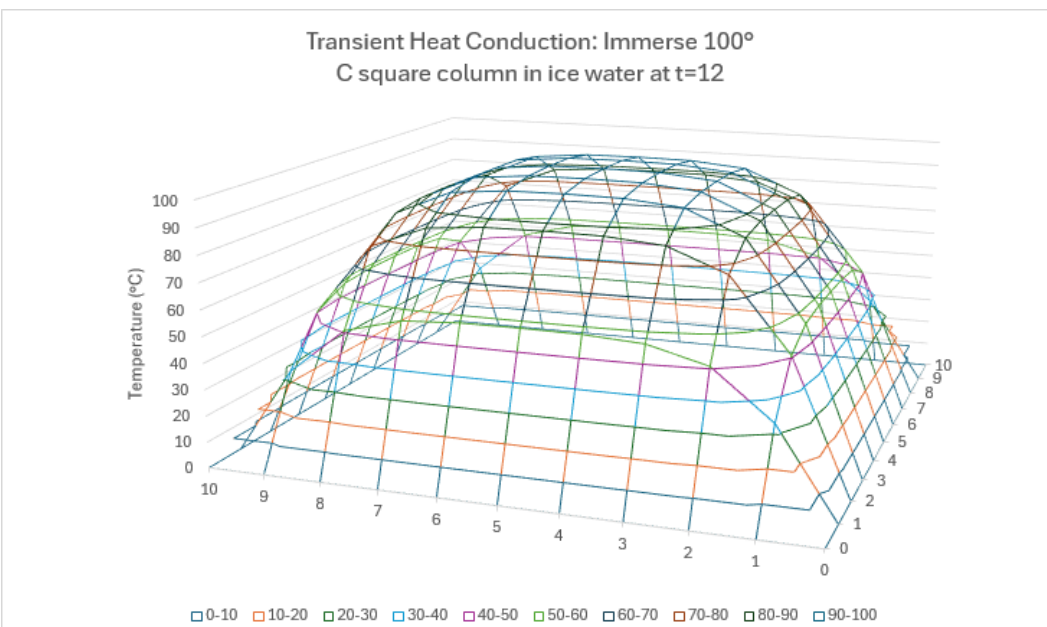

Figure 3: Transient state at t = 12.